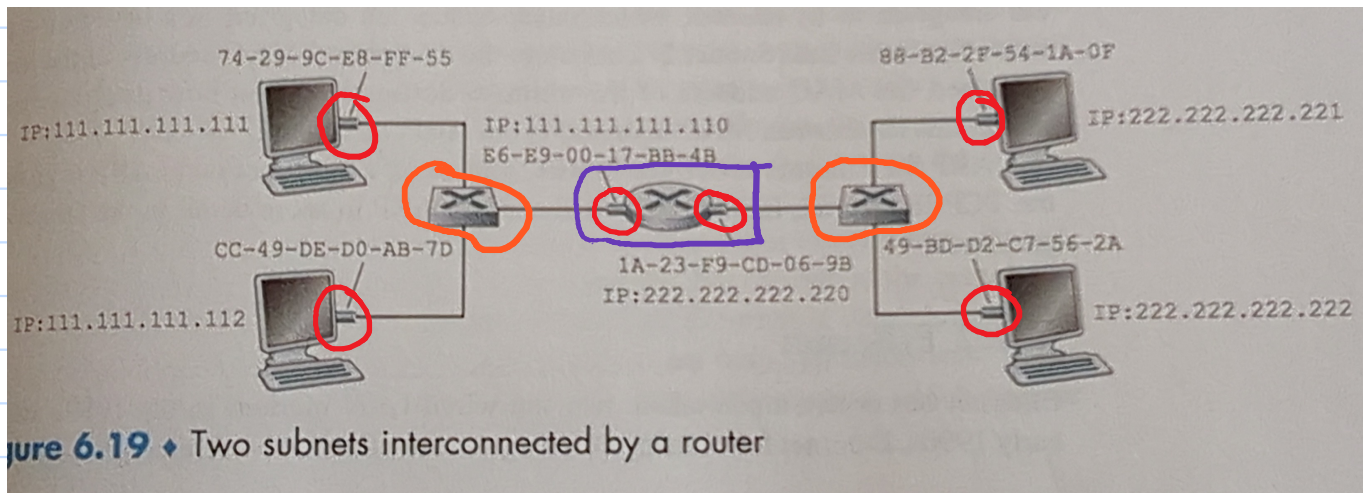


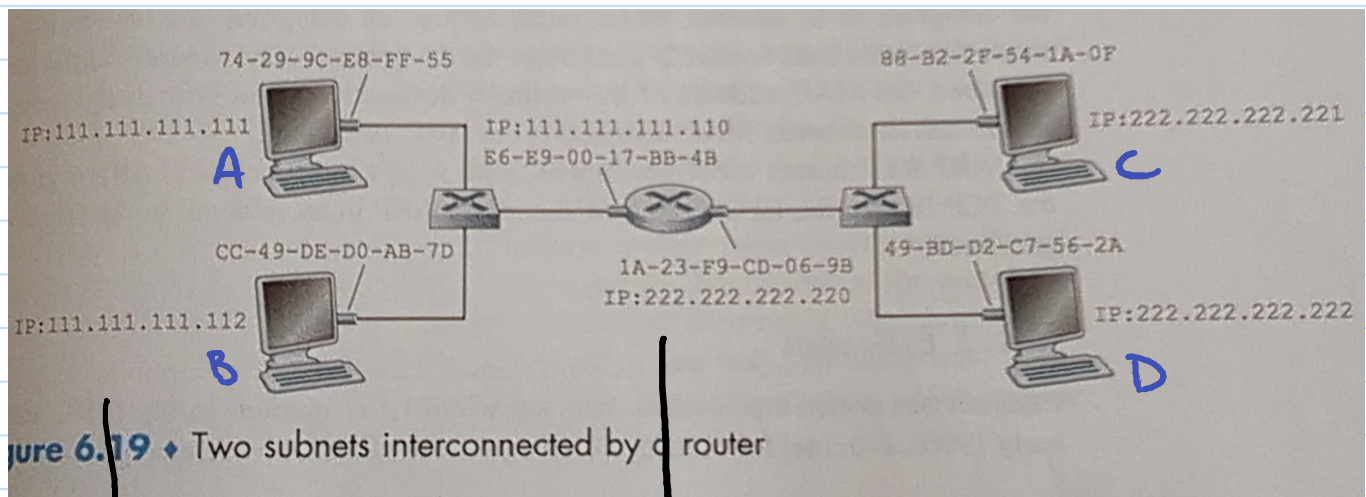


- - Network Interface (has IP and Mac Address)
- - Network Router (connects (sub)networks. Router IP)
- - Link-layer Switch (Forwards within network)

Address Resolution Protocol:

- (1) On Network? What MAC address has this IP? Send the packet there.
- (2) Off Network? What's the MAC address for the router to get this off network? Send the packet there.
- (3) Know the IP but not MAC? Broadcast a MAC Request ~ "Who Dis!?"

Ethernet/ARP	: Intra-Network (within Network)
IP	: Inter-Network (between networks)
TCP/UDP	: Intra-Application (between processes)



111.111.111.000/24

222.222.222.000/24

(1) Host A needs to send to B.

SRC IP & MAC?

DEST IP & MAC?

(2) Host A needs to send to D.

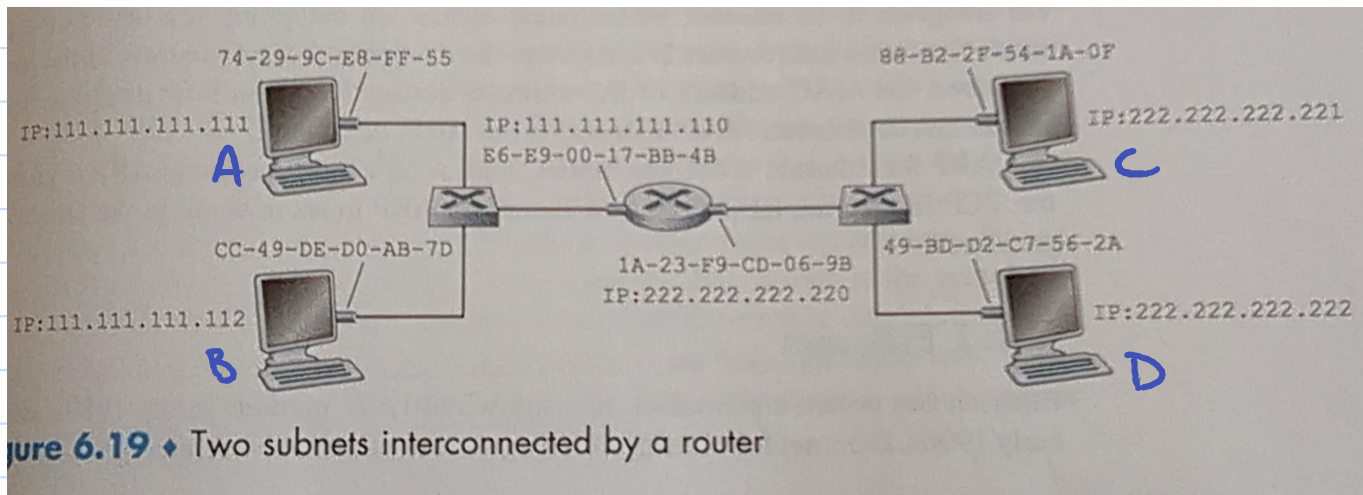
SRC IP & MAC?

DEST IP & MAC?

(3) Add Host E to 111.111.111.000/24.

Possible IP?

Possible MAC?



(1) C is new to 222.222.222.000/24. C wants to send to D. What happens?

(2) Now C wants to send to A. What happens?